# Pranay Kapoor

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## Education

**Manipal Institute of Technology** Bengaluru, India
*B.Tech in Computer Science (Honours)* *Jul 2023 – Present*

- CGPA: 8.68 / 10 | Scholars Scholarship recipient (top 9% of cohort)

## Technical Skills

**Languages**: Python, C, C++, Java
**Core CS**: Data Structures & Algorithms, OOP, Computer Architecture, Compiler Design
**Systems**: Processes, Threads, Memory Management, CUDA, MPI, Asyncio
**Data & Tools**: NumPy, Pandas, SQL, MongoDB, Git, Docker
**ML / AI**: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, OpenCV, MediaPipe

## Experience

**AI/ML Intern** Jun 2025
*BISAG-N, Ministry of Electronics & IT (MeitY), Government of India* *Gandhinagar, India*

- Built a full-stack medical imaging platform (React/Vite + Flask) for binary pneumonia detection and 4-class brain tumor classification (Glioma, Meningioma, Pituitary, Normal); MobileNet reached 90.06% accuracy and AUC-ROC of 0.97 on a 624-image chest X-ray test set.
- Benchmarked 5 architectures (baseline CNN, CNN + class weights, CNN + Squeeze-and-Excitation, ResNet50, MobileNet) for pneumonia and 3 for brain tumor (CNN, MobileNetV2, EfficientNetB0); implemented an ensemble voting mechanism with MobileNet tie-breaker fallback.
- Integrated Grad-CAM explainability pipeline surfacing pathological heatmap overlays on X-rays and MRIs; added OOD detection to reject non-medical inputs and flag low-confidence predictions (<85%) for manual review.

## Projects

**Apiro** | *Python, LLaMA 3.1 8B, HuggingFace, Shannon Entropy, Bioinformatics* | GitHub

- Belief graph engine using LLM token-entropy as a traversal heuristic over clinical pathways and gene-disease target spaces; high entropy flags under-explored decision boundaries, saturation self-terminates traversal. Calibration selected LLaMA 3.1 8B (T=0.7) over Mistral 7B across MedRAG, HPO, ClinVar, and OpenFDA evidence sources.

**NexaCred** | *Python, LightGBM, FastAPI, Docker* | GitHub | Live

- LightGBM credit scoring model on class-imbalanced DeFi wallet histories (F1 = 0.992, G-mean = 0.918); interest rates scaled to borrower risk score and served via FastAPI. Owned ML pipeline end-to-end within a group project.

**Flux Compiler (`fluxc`)** | *C99, GCC, GNU Make* | GitHub

- 3,972-line source-to-source compiler in C99 for a custom functional language (Flux) targeting optimized C output via GCC; supports pipe operators (`|>`), pattern matching, and recursion — compiled end-to-end without Flex/Bison.
- Hand-written lexer and recursive-descent parser with full operator precedence and scope management; multi-pass semantic analyzer validates type safety, function signatures, and recursion boundaries via scope-hierarchy tables.
- IR pass lowers AST to Three-Address Code (TAC), decoupling frontend from code generation; sub-millisecond transpile latency (<0.5 ms) and ∼1.0 MB peak heap measured on standard programs.

**Genetic Feature Selection & Hyperparameter Optimizer** | *Python, Scikit-learn, NumPy, Pandas* | GitHub | Live

- Genetic algorithm for joint feature selection and hyperparameter tuning across KNN, SVM, and Random Forest (fitness = 0.99×Acc + 0.01×Reduction); yielding +9.1% accuracy on Ionosphere and +1.6% on Breast Cancer, dropping up to 22 features over 10 generations.

**Face Emotion Recognition** | *PyTorch, ResNet-18, MediaPipe, Scikit-learn, OpenCV, Streamlit* | GitHub | Live

- Two-pipeline classifier: Pipeline 1 extracts MediaPipe landmarks → geometric ratios → Random Forest (<5 ms CPU inference); Pipeline 2 runs a grayscale ResNet-18 on FER-2013 with per-class loss weighting.
- Converted ResNet-18 input from 3-channel RGB to 1-channel grayscale, cutting tensor size by 66%; loss reweighting improved minority-class recall (Disgust, Fear) by 18%.

**Async API Caching Service (Arrodes)** | *Python, Asyncio, NLTK, BeautifulSoup4, Docker* | GitHub | Live

- Two-tier cache (in-memory + file-based) reduced single-request latency from 4.67 s to 76 ms (98% reduction), eliminating redundant network fetches and bypassing external rate limits.
- Offloaded blocking HTTP and parse tasks to asyncio thread pools; stress-tested under concurrent load — 9.5× higher throughput and 94% lower average latency versus uncached baseline.